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# Detecting fake reviews with supervised machine learning algorithms

## 用监督式机器学习算法检测虚假评论

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### ABSTRACT

This study provides an applicable methodological procedure applying Artificial Intelligence (AI)-based supervised Machine Learning (ML) algorithms in detecting fake reviews of online review platforms and identifies the best ML algorithm as well as the most critical fake review determinants for a given restaurant review dataset. Our empirical findings from analyzing 16 determinants (review-related, reviewer-related, and linguistic attributes) measured from over 43,000 online restaurant reviews reveal that among the seven ML algorithms, the random forest algorithm outperforms the other algorithms and, among the 16 review attributes, time distance is found to be the most important, followed by two linguistic (affective and cognitive cues) and two review-related attributes (review depth and structure). The present study contributes to the literature on fake online review detection, especially in the hospitality field and the body of knowledge on supervised ML algorithms.

### 摘要

本研究将基于人工智能(AI)的监督式机器学习(ML)算法应用于在线评论平台的虚假评论检测,并为给定的餐厅评论数据集识别出最佳的机器学习算法和最关键的虚假评论决定因素。我们分析了从43,000多条在线餐厅评论中得出的16个决定因素(评论相关,评论者相关和语言属性),实证结果表明,在7种机器学习算法中,随机森林算法优于其他算法,在16种评论属性中,时间距离是最重要的,其次是两个语言属性(情感和认知)和两个评论相关属性(评论深度和结构)。本研究对虚假在线评论检测方面的文献做出了贡献,特别是在酒店领域和监督式机器学习方面的知识体系。

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### KEYWORDS

Fake review; detection model development; online review platform; supervised machine learning; artificial intelligence; business intelligence

### 关键词

虚假评论; 检测模型开发; 在线评论平台; 监督式机器学习; 人工智能; 商业智能

## Introduction

User-generated online reviews have become increasingly popular as they can not only influence customers' product/service evaluations and purchase decisions (Flavián et al., 2016; Lee et al., 2017; Yang et al., 2017) but also affect service firms' business performance and service innovation (Lee et al., 2019; Öğüt and Onur Taş, 2012; Viglia et al., 2016). Online hotel and restaurant review platforms (e.g. TripAdvisor, Yelp) have been considered a reliable product/service information resource that has grown with the social engagement of consumers (Kesgin and Murthy, 2019), and have thus rapidly evolved over the past decade (Casaló et al., 2015).

However, recent studies have demonstrated that the credibility and transparency of these review platforms can be threatened by both businesses and customers posting fake reviews, leading to significant challenges for firms, customers, and review platforms themselves (Lo and Yao, 2019; Luca and Zervas, 2016). For example, 67 percent of U.S. consumers believe that fake online reviews are problematic but difficult to identify (Escobar, 2021). Spotting fake reviews has become a major concern in online transactions and purchase decision-making, making it necessary to address this issue by having online review platforms filter them in order to preserve consumer confidence (Escobar, 2021; Witts, 2019). In the hospitality industry, in particular, TripAdvisor and Yelp have made significant efforts to detect and remove fake reviews (Post Modern Marketing, 2021; Witts, 2019), but 7.1 percent of Yelp reviews and 5.2 percent of TripAdvisor reviews are still found to be counterfeit (Escobar, 2021). There is thus still an urgent call for developing machine learning (ML) and artificial intelligence (AI) tools to reduce online review fraud, which would lead to the enhancement of customer trust (Escobar, 2021; Kandampully et al., 2022).

A fake online review is an online fraud review consisting of either an unwarranted positive comment regarding a product/service for a promotion or an unjustified negative response aimed at damaging the reputation of a product/service (Jindal and Liu, 2008). As detecting these fake reviews is a difficult task due to the absence of physical cues (e.g. fidgeting, a lack of eye contact) (Baker and Kim, 2019; Yoo and Gretzel, 2009), prior research has focused on identifying determinant factors that would make review fraud more likely (e.g. Luca and Zervas, 2016; Malbon, 2013) as well as factors that detect fake reviews based on linguistic styles using manual identification and human coding (e.g. Li et al., 2020; Plotkina et al., 2020). Review- or reviewer-specific information can also be used to detect fake reviews (e.g. Bradner and Mark, 2002; Li et al., 2020; Zhou and Guo, 2017).

ML is the umbrella term for AI and refers to intelligence demonstrated by machines or computer systems in contrast to the natural intelligence displayed by humans. AI enables us to enhance the performance of mechanical, analytical, intuitive, and empathetic tasks in the service setting (Huang and Rust, 2018). Online review platforms and service providers must manage and analyze online review data that is constantly growing in volume, variety, and velocity (Singh et al., 2017). Given that AI-based ML can be an essential method of processing and analyzing online reviews, recent studies in the service industry, computer science, and information systems (IS) fields have used various ML techniques along with web-scraping, text-mining, and sentiment analysis techniques to analyze the impact of sentiments embedded in online reviews (Buzova et al., 2019; Kesgin and

Murthy, 2019; Lee et al., 2021), predict firm bankruptcy (Kim, 2011), and predict or detect fake online reviews (Wu et al., 2020). A few studies have recently started utilizing these ML and text-mining techniques in the service literature to propose a methodological procedure of predicting fake online reviews or developing ML-based fake review detection models (e.g. Martinez-Torres and Toral, 2019).

Overall, we have identified the following two research gaps regarding (1) the literature on fake review determinants detection in the service industry and (2) ML-based fake review detection techniques. As shown in Table 1, there is still little research exploring fake reviews in the service industry in the literature on fake review determinants. In particular, AI-based ML techniques have not been actively applied in the hospitality service field, which is described in the following section. Thus, by comparing the relative importance of various characteristics of restaurant reviews that predict the likelihood of fake

**Table 1.** Determinants of fake online reviews in existing studies.

| Feature                                      | Description                                                                                       | Research method                                       | Prior research                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------|
| <i>Category: Linguistic attributes</i>       |                                                                                                   |                                                       |                                                          |
| Affective Cues                               | The extent to which (ETW, hereinafter) how many affective words are included <i>in the review</i> | Algorithm development                                 | Ott et al. (2011)                                        |
| Cognitive Cues                               | ETW how many cognitive words are included                                                         | Scenario-based interview                              | Matsumoto and Hwang (2015)                               |
| Social Cues                                  | ETW how many social words are included                                                            | Social network analysis                               | Brown and Reingen (1987)                                 |
| Perceptual Cues                              | ETW how many perceptual words are included                                                        | Scenario-based interview                              | Matsumoto and Hwang (2015)                               |
| Motion Words                                 | ETW how many motion words are included                                                            | Text mining (LIWC software)                           | Tausczik and Pennebaker (2010)                           |
| Psychological Distancing                     | ETW how much distance is between the reviewer and the reader of the review                        | Text mining (LIWC software)                           | Tausczik and Pennebaker (2010); Toma and D'Angelo (2015) |
| <i>Category: Review-related attributes</i>   |                                                                                                   |                                                       |                                                          |
| Review Depth                                 | ETW how many words are used                                                                       | Unsupervised hierarchical approach                    | Kumar et al. (2019)                                      |
| Review Structure                             | ETW how many words per sentence are used                                                          | Supervised machine learning for algorithm development | Banerjee et al. (2015)                                   |
| Writing Style                                | ETW how focused the review is about present/future                                                | Supervised machine learning for algorithm development | Banerjee et al. (2015); Tausczik and Pennebaker (2010)   |
| Review Star Rating                           | ETW what rating the review receives                                                               | Text-mining and random forest-based classifiers       | Ghose and Ipeiritos (2010)                               |
| Review Extremity                             | ETW how extreme a review rating is                                                                | Unsupervised hierarchical approach                    | Kumar et al. (2019)                                      |
| Number of Photos                             | ETW how many photos are uploaded                                                                  | Image mining                                          | Li et al. (2020)                                         |
| <i>Category: Reviewer-related attributes</i> |                                                                                                   |                                                       |                                                          |
| Reviewer Reputation                          | ETW how many reviewer's friends are on Yelp.com                                                   | Bootstrap analysis                                    | Zhu et al. (2014)                                        |
| Reviewer Experience                          | ETW how many reviews a reviewer had posted before the sampled review                              | Text mining, a regression discontinuity approach      | Luca (2016)                                              |
| Time Distance                                | ETW how much time had passed between the first review and the current review                      | Text mining and negative binomial regression          | Zhou and Guo (2017)                                      |
| Reviewer Identity Deception                  | Fake identity of peer or expert reviewers                                                         | Survey                                                | Akhtar et al. (2019)                                     |
| Reviewer Location                            | Whether or not a reviewer's location is the target restaurant's location.                         | Text mining and Tobit regression                      | Liu and Park (2015)                                      |

reviews, this study contributes to the literature on fake online reviews with a focus on the hospitality service context.

For ML-based fake review detection techniques, the academic fields of computer science and IS have actively applied ML (Wu et al., 2020), using both supervised (e.g. Banerjee et al., 2015; Zhang et al., 2016) and unsupervised ML techniques (e.g. Kumar et al., 2019) to propose (or fine-tune) the best fake review detection algorithms and compare the characteristics of fake vs. genuine reviews in terms of various features embedded in online reviews: content- (e.g. linguistic characteristics or sentiments), review- (e.g. star rating or helpfulness rating), and reviewer-related (e.g. reviewers' level of experience/expert) features. However, in the academic field studying the hospitality service, the research on fake reviews has focused on the empirical test of relationships between review characteristics and the occurrence of fake reviews or between the event of deceptive reviews and their consequences, applying statistical techniques other than ML, such as the qualitative comparative analysis (QCA), *t*-test, regression, and structural equation modeling (SEM) (e.g. Akhtar et al., 2019; Lee et al., 2018; Li et al., 2020). However, to the best of our knowledge, in the field of hospitality service, little work has been done to identify the most effective fake review detection method by comparing multiple ML-algorithms integrating and using all three groups of predictor variables (i.e. linguistic characteristics of reviews, reviewer- and review-related non-content characteristics) extracted from a large online review dataset.

To address these research gaps, this research aims to provide a rigorous understanding of AI-based ML applications by merging the two research streams, including fake review detection (the literature) and ML algorithms (the methodology). In particular, this study first attempts to build a fake review prediction model by identifying and integrating key determinants proposed and validated in the IS, marketing, and computer science literature as well as recent hospitality service research, resulting in various review characteristics that can detect fake reviews in the restaurant setting. Using these potential fake-review determinants, we provide researchers and practitioners with a methodological procedure for (1) collecting sample online product/service reviews, (2) measuring possible determinants of fake reviews automatically with a set of formulas, tools, and techniques, such as web-scraping, text-mining, and sentiment analysis, (3) running and comparing the effectiveness of classification models with various supervised ML-algorithms, and (4) finally, identifying the best set of determinants based on the best performing model rather than testing proposed hypotheses in order to improve the predictability and credibility of online hospitality service review platforms. To achieve these research purposes, this study identifies sixteen possible fake-review determinants, including review- and reviewer-related attributes as well as linguistic characteristics of the reviews, and proposes seven supervised ML algorithms to predict fake online reviews, followed by an additional test to identify the best fake review determinants applied to the best performing model suggested by our rigorous model validation process.

Compared to a recent study on fake review detection in the hospitality service field (Martinez-Torres and Toral, 2019), the present study is one of the first attempts to integrate critical factors affecting the occurrence of fake reviews drawn from extant literature and propose an ML-based fake review detection/prediction model. Therefore, our study contributes to the literature on fake online review detection, especially in the hospitality service field and the body of knowledge on supervised-ML algorithms. Additionally, our

results will help practitioners enhance their understanding of the factors influencing the occurrence of fake reviews and eventually improve the credibility of online review communities.

## Literature review and research background

### *Fake online review research in the hospitality service industry*

Fake online reviews are considered a deviant online behavior (of both customers or businesses) and refer to intentionally deceitful (fraudulent) online reviews describing either unjustified negative opinions about a target business to harm its reputation or unwarranted positive opinions about the business to enhance its reputation or performance (Jindal and Liu, 2008; Li et al., 2020). Detecting fake reviews is a difficult task due to their anonymous nature of reviews and their lack of physical cues (Bradley et al., 2015; Yoo and Gretzel, 2009). Nonetheless, the task is crucial for the credibility of online review communities in the hospitality service industry (Lo and Yao, 2019). Prior research on fake online reviews in the hospitality service sector has focused on three major types of review attributes that help detect fake online reviews: linguistic, review-related, and reviewer-related attributes. We thus reviewed the hospitality service literature to prepare a list of potential determinants for fake review occurrence to be added to our fake review prediction model development. These potential determinants are categorized into the three types mentioned above: linguistic, reviewer-related, and review-related attributes. Table 1 lists the names and descriptions of the attributes and references.

***Linguistic attributes as determinants for fake reviews.*** Extant studies have focused on identifying the determinant factors of review fraud (Luca and Zervas, 2016; Malbon, 2013) and detecting fake reviews based on linguistic styles (Jia et al., 2018; Li et al., 2020; Plotkina et al., 2020; Shan et al., 2021). The reason that linguistic analysis can be used to detect fake reviews is that different linguistic styles are likely to be used when telling the truth versus a lie (Pennebaker et al., 2003), and these differences in linguistic styles can significantly influence the understanding and evaluation of online reviews (Toma and Hancock, 2012; Van Laer et al., 2019). Some of the specific linguistic cues that have been examined in extant studies on the detection of fake online reviews include affective cues (e.g. Hajek et al., 2020; Li et al., 2020; Ott et al., 2011), cognitive cues (e.g. Banerjee et al., 2015; Matsumoto and Hwang, 2015; Wahyuni and Djunaidy, 2016), social cues (e.g. Barbado et al., 2019), perceptual cues (e.g. Li et al., 2020; Matsumoto and Hwang, 2015), motion words (e.g. Tausczik and Pennebaker, 2010), pronouns (such as subjectives and objectives) (e.g. Alsubari et al., 2020), counts of verbs, adverbs, nouns (e.g. Alsubari et al., 2020), redundant terms (e.g. Abri et al., 2020), and psychological distancing (e.g. Tausczik and Pennebaker, 2010; Toma and D'Angelo, 2015). Therefore, linguistic styles have been considered the critical determinant for fake review detection.

***Review-related attributes as determinants for fake reviews.*** Review- and reviewer-related characteristics have been found to be significantly associated with review helpfulness, consumers' information adoption, and purchase decision (Siddiqui et al., 2020), and they have also been used in studies to detect fake reviews (Zhang et al., 2016). A review is unarguably the major element of fake reviews; extant studies have thus extensively examined review-related characteristics as major fake review determinants. For review

characteristics, review depth (or length) (e.g. Kumar et al., 2019), review structure, such as the level of detail in a review (e.g. Banerjee et al., 2015), review timing (e.g. Moon et al., 2021), writing style in terms of tenses (e.g. Banerjee et al., 2015), argument structure and contextual embedding (e.g. Ansari and Gupta, 2021), review star rating (e.g. Ghose and Ipeirotis, 2011; Luca and Zervas, 2016), review rating extremity (Kumar et al., 2019; Luca and Zervas, 2016), number of photos (e.g. Li et al., 2020; Shan et al., 2021), review vote and update counts (e.g. Shan et al., 2021), review ratings (e.g. Zhang et al., 2016), review details, such as 'spatial details' (e.g. Li et al., 2013; Ott et al., 2011), and review honesty and readability value (e.g. Wahyuni and Djunaidy, 2016; Zhang et al., 2016) have been used by extant studies for the detection of fake reviews. Overall, when considering the hospitality service literature, we believe that review-related attributes should be added to the development of our fake review detection model as determinants for fake review detection.

**Reviewer-related attributes as determinants for fake reviews.** Reviewer characteristics have also been incorporated into fake review detection models. Motivated by the desire to increase reliability and the feeling of connection with reviewers, reviewers' information and their interactive activities with readers are made available (Ansari and Gupta, 2021). For the reviewer characteristics, a reviewer's reputation (i.e. elite reviewer status, favorable votes count, trustiness value, and number of friends and followers) and experience (i.e. the time since the reviewer wrote their first review and the length of time they spent as a member in the review community) (e.g. Luca, 2016; Shan et al., 2021; Wahyuni and Djunaidy, 2016; Zhang et al., 2016; Zhu et al., 2014), time distance (e.g. Zhou and Guo, 2017), reviewer location (e.g. Deng et al., 2018; Liu and Park, 2015), reviewers' identity deception, disclosure, avatar used (e.g. Akhtar et al., 2019; Munzel, 2016; Zhang et al., 2016), reviewers' review-writing behaviors (i.e. number of reviews written and review writing frequency), and nonverbal behaviors of reviewers (e.g. number of postings each day, activity windows, check-in counts, and interactions with others in the review community) (e.g. Mukherjee et al., 2013a; Mukherjee et al., 2013b) have been used by extant studies as possible features to detect fake reviews. Thus, reviewers' rich, reliable, and diversified characteristics are chosen to be the determinants for fake reviews in our fake review detection model.

### *ML approach for fake reviews*

In general, ML is defined as a 'field of study that gives computers the ability to learn without being explicitly programmed' (Munoz, 2014, p. 1). However, this term often refers to the ML algorithms that apply statistical techniques to determine meaningful patterns using large datasets and allow technological systems to learn from data, identify patterns, and make decisions with minimal human interventions (Hao, 2018; SAS, 2020). As the technological advancement of ML algorithms is rapidly increasing their applications in many customer services in various industries (Flavián and Casaló, 2021; Huang and Philp, 2021), some ethical issues with these AI-based applications have been raised (Belk, 2021). Just like ML algorithms are used to detect fake reviews, it is also possible to generate fake reviews with AI (Goldschmidt, 2019), which may result in seriously misleading consumer information and service interactions (between human customers and AI-powered frontline non-human workers) (Castillo et al., 2021). These

possible ethical issues further highlight the importance of a rigorous methodological approach for detecting maliciously generated fake written content.

***Supervised ML algorithms for fake review detection.*** Supervised ML is often preferred for the detection of fake reviews (or spam reviews) because the outcome of a fake review is binary (with authentic vs. fake reviews), and obtaining a previously labeled training dataset (e.g. in the case of Yelp.com) is relatively easier (Zhang et al., 2016). Therefore, quite a few extant studies have used supervised ML to detect or compare the different impacts of the factors influencing fake reviews and have reported the performance of various supervised ML methods. Wu et al. (2020)'s literature review contains a list of extant studies that have used ML to detect fake (or spam) reviews with the list of applied ML algorithms. Based on Wu et al. (2020), support vector machines (SVM), neural network, naïve Bayes, logistic regression, k-nearest-neighbor (k-NN), decision (classification) tree, and random forest algorithms are the supervised ML techniques thus far used by extant studies to detect fake reviews, and SVM, neural network, naïve Bayes, and logistic regression are used more frequently than the other three algorithms. Therefore, this study focuses on supervised ML algorithms and compares the performance of diverse methods to propose the best model for fake review detection using a large dataset.

## Research methodology

### *Business intelligence and supervised ML procedures*

To accomplish our research objectives, we used business intelligence (BI) and AI techniques. These techniques include data mining, text analysis, and supervised ML approaches. To develop the classification model of fake online reviews, we adopted and integrated BI and supervised ML approaches because ML outperforms other techniques in fake review detection (Wu et al., 2020; Zhang et al., 2016). ML algorithms are typically classified into three types – supervised, unsupervised, and semi-supervised – based on the existence and use of a correctly labeled dataset to train the machine to achieve a certain level of model performance (Crawford et al., 2015). Supervised ML uses a set of labeled (for correct true-positive and true-negative values of the outcome variable) data (i.e. training dataset) to train the model, validates it with another set of data (i.e. validation dataset), and predicts the outcomes of the model with testing data that consist of a group of predictor variables. Unsupervised ML does not use a training dataset but can determine unseen or unexpected patterns using unlabeled data. Semi-supervised ML uses a mix of both supervised and unsupervised techniques in that it commonly handles some labeled but more unlabeled data to train the model to predict its outcomes with testing data (Crawford et al., 2015; Lighthart et al., 2021). In this study, we used supervised ML as the primary goal of this study is to predict fake online reviews.

We conducted our study as follows. First, key determinants of fake reviews and ML algorithms were identified by reviewing the extant literature. Second, online restaurant reviews of the top 100 U.S. restaurants were collected from Yelp.com and then analyzed by BI techniques. The prepared dataset was randomly split into a training dataset (80%) and a validation dataset (20%) for ML model development. Then, seven ML algorithms were applied to develop the best classification model detecting fake reviews based on

the evaluation metrics. Lastly, an additional diagnostic test was performed to identify the relative importance of fake review determinants. Figure 1 depicts BI and supervised ML procedures employed in this study.

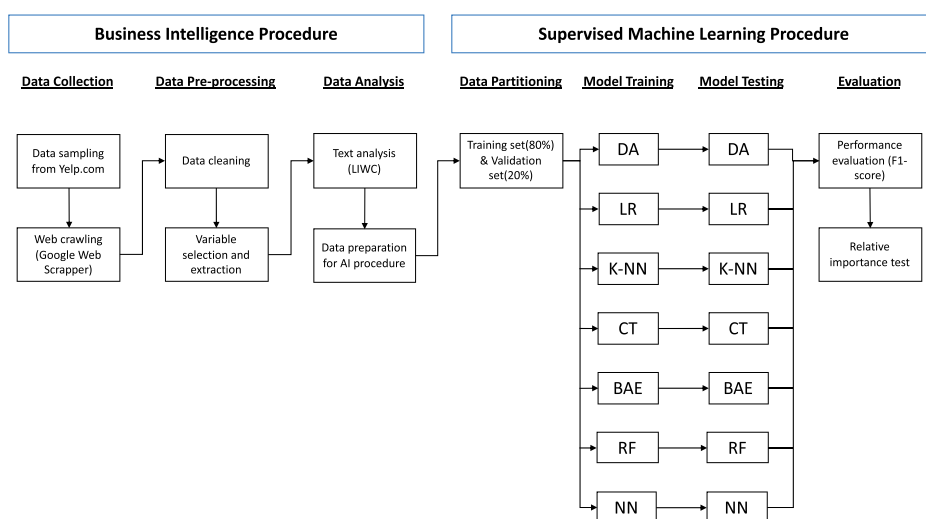
### Data collection

As the first step, online restaurant review data were collected from the 'Top 100 Places to Eat for 2017 in the U.S.' list on Yelp.com. Yelp.com is one of the most popular online review platforms in the hospitality context (Li et al., 2020; Luca, 2016) and has over 38 million restaurant reviews (Yelp, 2020). Yelp.com is reputed to be a reliable online review platform owing to its advanced system of filtering out fake reviews (Yang et al., 2017) and is thus considered a qualified data source (e.g. Lee et al., 2021). We selected the top 100 U.S. restaurants because they represent the U.S. restaurant market and reduce location bias (Li et al., 2020). More importantly, the selected restaurant reviews were marked as fake and authentic by Yelp.com's fake review screening algorithms, which is critical to developing a classification model of fake online reviews. Google Web Scraper was utilized to collect 43,496 reviews, which contained 36,742 authentic reviews and 6,754 fake reviews.

### Measurements

BI refers to the techniques or methodologies (e.g. data mining and text mining) that collect and analyze critical business data (Chen et al., 2012). This advanced technique has been used in the hospitality service field to uncover hidden patterns and new knowledge and test proposed hypotheses and theories (Lee et al., 2017; Lee et al., 2019; Xiang et al., 2017; Xiang et al., 2015).

For data analysis, we pre-processed the initial dataset by checking for non-ASCII characters, reviews without actual text, and other critical information (e.g. review- and



**Figure 1.** Business intelligence and supervised machine learning procedures.

reviewer-specific information) based on Table 1. We then used Linguistic Inquiry and Word Count 2015 (LIWC2015) to measure linguistic attributes such as affective cues, cognitive cues, social cues, perceptual cues, motion words, and psychological distancing (Pennebaker et al., 2015). LIWC2015 has been extensively used in psychology, communication (Tausczik and Pennebaker, 2010), and hospitality settings (Lee et al., 2019; Li et al., 2020). This renowned text analysis program consists of two main components, such as the text analysis processing module and the pre-developed dictionary (i.e. the collection of words in 72 linguistic dimensions). The processing component processed the pre-processed review text files and measured the linguistic and psychological dimensions based on the LIWC2015 dictionary.

The present study measured six linguistic features related to fake review detection, as shown in Table 2. To measure the linguistic and review-related features, we utilized the

**Table 2.** Measurements and summary statistics of variables.

| Measurement for variables                                                                                                                                                                                                                                     | Mean   | S.D.   | Min | Max   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-----|-------|
| <i>Fake Reviews (Dependent Variable) (Luca and Zervas, 2016)</i>                                                                                                                                                                                              |        |        |     |       |
| Whether or not a review is perceived by Yelp as fake (0: authentic review; 1: fake review)                                                                                                                                                                    | 0.16   | 0.36   | 0   | 1     |
| <i>Affective Cues (Ott et al., 2011)</i>                                                                                                                                                                                                                      |        |        |     |       |
| The proportion of affective words [1,393 Representative Words in LIWC2015 (e.g. happy, cried, love, nice, hurt, ugly, and nasty): (1) positive emotion words (2) negative emotion words]                                                                      | 9.78   | 5.93   | 0   | 100   |
| <i>Cognitive Cues (Matsumoto and Hwang, 2015)</i>                                                                                                                                                                                                             |        |        |     |       |
| The proportion of cognitive words [797 Representative Words in LIWC2015 (e.g. cause, know, think, because, should, & but ...): (1) insight words (2) causation words (3) discrepancy words (4) tentative words (5) certainty words (6) differentiation words] | 9.16   | 4.93   | 0   | 80.95 |
| <i>Social Cues (Brown and Reingen, 1987)</i>                                                                                                                                                                                                                  |        |        |     |       |
| The proportion of social words [756 Representative Words in LIWC2015 (e.g. mate, talk, they, buddy, and together): family, friends, female/male references, etc.]                                                                                             | 8.15   | 5.21   | 0   | 100   |
| <i>Perceptual Cues (Matsumoto and Hwang, 2015)</i>                                                                                                                                                                                                            |        |        |     |       |
| The proportion of perceptual words [436 Representative Words in LIWC2015 (e.g. look, heard, feeling, saw, listen, and feels): see, hear, feel, etc.]                                                                                                          | 2.98   | 2.96   | 0   | 100   |
| <i>Motion words (Tausczik and Pennebaker, 2010)</i>                                                                                                                                                                                                           |        |        |     |       |
| The proportion of motion words in LIWC2015 (e.g. arrive, car, go, etc.)                                                                                                                                                                                       | 1.95   | 2.17   | 0   | 100   |
| <i>Psychological Distancing (Tausczik and Pennebaker, 2010; Toma and D'Angelo, 2015)</i>                                                                                                                                                                      |        |        |     |       |
| The proportion of first-person pronouns in LIWC2015 (e.g. I, me mine, we, us, our, etc.)                                                                                                                                                                      | 7.32   | 4.10   | 0   | 33.33 |
| <i>Review Depth; Review Structure (Kumar et al., 2019; Banerjee et al., 2015)</i>                                                                                                                                                                             |        |        |     |       |
| The number of words used in the review (word count)                                                                                                                                                                                                           | 78.99  | 79.98  | 1   | 970   |
| The number of words per sentence in the review (words/sentence)                                                                                                                                                                                               | 11.55  | 6.87   | 1   | 527   |
| <i>Writing Style (Banerjee et al., 2015; Tausczik and Pennebaker, 2010)</i>                                                                                                                                                                                   |        |        |     |       |
| The proportion of present/future focus (present and future focus words) in LIWC2015 (e.g. today, is, now, may, will, soon, etc.)                                                                                                                              | 10.06  | 5.58   | 0   | 100   |
| <i>Review Star Rating; Review extremity (Ghose and Ipeirotis, 2010; Kumar et al., 2019)</i>                                                                                                                                                                   |        |        |     |       |
| The rating of the review based on the 5-star scale                                                                                                                                                                                                            | 4.78   | 0.66   | 1   | 5     |
| The extremity of a review rating (the quadratic term of the rating)                                                                                                                                                                                           | 23.24  | 4.74   | 1   | 25    |
| <i>Number of Photos (Li et al., 2020)</i>                                                                                                                                                                                                                     |        |        |     |       |
| Number of photos in the review                                                                                                                                                                                                                                | 0.33   | 0.80   | 0   | 11    |
| <i>Reviewer Reputation (Zhu et al., 2014)</i>                                                                                                                                                                                                                 |        |        |     |       |
| The number of reviewer's friends on Yelp.com                                                                                                                                                                                                                  | 59.88  | 223.50 | 0   | 13622 |
| <i>Reviewer Experience (Luca, 2016)</i>                                                                                                                                                                                                                       |        |        |     |       |
| The number of reviews a reviewer had posted before the sample review                                                                                                                                                                                          | 54.46  | 199.50 | 1   | 11800 |
| <i>Time Distance (Zhou and Guo, 2017)</i>                                                                                                                                                                                                                     |        |        |     |       |
| The time difference between the first review and the current review                                                                                                                                                                                           | 831.35 | 722.23 | 0   | 3940  |
| <i>Reviewer Location (Liu and Park, 2015)</i>                                                                                                                                                                                                                 |        |        |     |       |
| Whether or not a reviewer's location is the target restaurant's location (0: the same location; 1: otherwise)                                                                                                                                                 | 0.05   | 0.22   | 0   | 1     |

following linguistic categorization types in LIWC2015: psychological process (i.e. affective, social, cognitive, social, perceptual cues), relativity (i.e. motion words), and language dimensions (i.e. first-person pronouns, present focus words, future focus words, word count, and the number of words per sentence). For a dependent variable, we used the annotated information indicating whether or not a review is fake. Table 2 presents the operationalization of the variables and their descriptive statistics.

### *Fake review detection model development*

The present study proposes the fake review detection model by employing the popular supervised ML algorithms, comparing their predicting performance, and identifying the relative importance of fake review determinants. To accomplish these goals, we used an analytic data mining tool called XLMiner, a comprehensive program providing data mining, text mining, and predictive analytics (Frontline Systems, 2020; Shmueli et al., 2016). XLMiner provides various statistics and business analytics techniques, including prediction, classification, optimization, forecasting, and simulation. Specifically, it offers both supervised ML algorithms (e.g. neural networks, classification and regression trees, and regressions) and unsupervised ML algorithms (e.g. association rules, sequential pattern, and principal components analysis).

As shown in Figure 1, the supervised ML procedure consists of data partitioning, model training, model testing, and performance evaluation. To train and test the supervised ML models, we created a training dataset (80%) and a validation dataset (20%) by randomly splitting the dataset prepared in the BI procedure as suggested by prior ML research (e.g. Akay, 2009; Polat and Güneş, 2007). The training dataset is the portion of the data used to fit a model, and the validation dataset is the portion of the data used to assess how well the model fits predicted models and to select the best model among the seven models using different ML algorithms (Shmueli et al., 2016). The current study selected seven supervised ML-algorithms, namely, discriminant analysis, logistic regression, k-nearest-neighbor (k-NN), classification tree, bootstrap aggregating ensembles (BAE), random forest, and neural network because they are the most frequently used classification techniques (Wu et al., 2020; Zhang et al., 2016) in the IS and computer science literature and are also available in XLMiner. To determine the best model, seven different ML classifiers were trained and evaluated based on the performance metrics (e.g. precision, specificity, sensitivity, F1-score, and overall error rate).

Among the various supervised ML algorithms (Wu et al., 2020), we first chose discriminant analysis because it is a more traditional (widely known) and simpler method than SVM, which requires more complex and sometimes inefficient model validation processes (Clemmensen et al., 2011; Ståpor, 2016). Although evidence proves that SVM performs better than discriminant analysis in some cases of classifications (e.g. where the decision boundary is non-linear or multi-dimensional) (Clemmensen et al., 2011; Zhang et al., 2016), discriminant analysis typically performs as well as SVM where the classification question is relatively simple (e.g. binary classification such as spam/non-spam reviews or default/non-default credit scorings) as is the case in this study. Logistic regression, as the extension of multiple linear regressions, can be used when the dependent variable is categorical (Shmueli et al., 2016). Logistic regression was selected because it can be used to clarify a new observation in the dataset where the class is binary and known.

It is thus commonly used to detect fake reviews and spam messages (e.g. Banerjee et al., 2015; Liu et al., 2019). Random forest and BAE are ensemble-based classifiers that classify the outcome variable based on a classification tree. The classes are generated by individual trees using a subset of randomly restricted and selected predictors (Zhang et al., 2016). The classification tree is a traditional classification method that uses a tree-like graph through recursive partitioning and pruning procedures (Shmueli et al., 2016) and performs well in the context of online deceptive text detection (Ball and Elworthy, 2014). In addition, K-NN and neural networks have been extensively used in situations in which researchers attempt to predict and classify binary outcomes related to spam or deceptive review detection (Wu et al., 2020).

As suggested by previous ML literature (e.g. Martinez-Torress and Toral, 2019; Shmueli et al., 2016; Zhang et al., 2016), the performance of the seven fake review classification models were evaluated by the following evaluation metrics as shown in Table 3: precision, specificity, sensitivity (recall), F1-Score, and overall error rate. While precision is the proportion of total fake reviews identified by each algorithm, sensitivity is the proportion of fake reviews correctly classified in the validation dataset. Following these measures, we computed F1-scores of each classification model as described in Equation (1). The F1-score is a harmonic mean of precision and sensitivity and can be used for classification model comparisons (Martinez-Torress and Toral, 2019; Shmueli et al., 2016; Zhang et al., 2016).

$$F1 - score = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity} \quad (1)$$

## Results

To address our proposed research objectives, the seven supervised ML-algorithms (i.e. discriminant analysis, logistic regression, k-NN, classification tree, BAE, random forest, and neural network) were implemented and tested using XLMiner. As suggested by prior ML research (Martinez-Torres and Toral, 2019; Wolpert and Macready, 1997), we attempted to employ multiple algorithms because one algorithm is not guaranteed to constantly perform better than the others, as its performance can vary depending on various factors and conditions (Shmueli et al., 2016).

80% of the original dataset was randomly selected and trained for initial model development, and the remaining 20% was used to evaluate and measure performance metrics as the validation dataset. The performance of each ML algorithm was compared and

**Table 3.** Evaluation results of the seven proposed predictive models (80:20 sampling).

| Model                 | Classification performance measures |              |                      |              |                    |
|-----------------------|-------------------------------------|--------------|----------------------|--------------|--------------------|
|                       | Precision                           | Specificity  | Sensitivity (Recall) | F1-Score     | Overall Error Rate |
| Discriminant analysis | 0.951                               | 0.997        | 0.311                | 0.469        | 11.13%             |
| Logistic regression   | 0.964                               | 0.998        | 0.315                | 0.475        | 10.99%             |
| k-NN                  | 0.299                               | 0.885        | 0.263                | 0.280        | 21.36%             |
| Classification tree   | 0.484                               | 0.899        | 0.504                | 0.494        | 16.31%             |
| BAE                   | 0.694                               | 0.961        | 0.473                | 0.563        | 11.61%             |
| <b>Random forest</b>  | <b>0.724</b>                        | <b>0.967</b> | <b>0.462</b>         | <b>0.564</b> | <b>11.27%</b>      |
| Neural network        | 0.161                               | 0.028        | 0.993                | 0.277        | 81.96%             |

evaluated based on the performance metrics as described in the previous section. [Table 3](#) presents detailed performance metrics, including precision, specificity, sensitivity (recall), F1-Scores, and overall error rates (misclassification error). According to the F1-Score, the random forest algorithm (F1-score = 0.564) outperformed the other ML algorithms. Although the BAE algorithm (F1-score = 0.563) performed similarly, other performance measures were lower than those of the random forest algorithm. Thus, the random forest algorithm-based classification model outperforms other models in detecting fake reviews. It is noteworthy that the neural network-based model worked poorly in the current dataset because neural network algorithm performance can be significantly influenced by the sample size (Markham and Rakes, 1998).

Subsequently, to analyze robustness, we selected the best performing random forest model and analyzed the relative importance of individual determinants for fake review classification, which has been rarely examined in the previous literature (Zhang et al., 2016). Among the 16 fake review determinants, time distance was found to be the most important feature, followed by linguistic attributes (i.e. affective cues, cognitive cues) and review attributes (i.e. review depth, review structure). Interestingly, the number of photos and the reviewer's location were identified as the least important features among others. [Table 4](#) presents the rank and the relative importance of each of the features of fake reviews.

## Discussion

### Research findings

The present study aims to identify the factors predicting fake reviews, compare their relative importance, and suggest the best way to predict the likelihood that a review is classified as a fake review using the identified factors in the hospitality context. To address these objectives, this study uses the ML-based classification model. Put simply, a classification model is a statistical technique to predict a dichotomous dependent variable, such as 'genuine vs. fake' reviews in the context of the current study or 'non-defect vs. defect' in a quality control context. A supervised ML algorithm can be added to the

**Table 4.** The relative importance of fake review determinants.

| Relative importance rank | Feature                  | Importance score |
|--------------------------|--------------------------|------------------|
| 1                        | Time Distance            | 14.190           |
| 2                        | Affective Cues           | 12.827           |
| 3                        | Review Depth             | 12.670           |
| 4                        | Review Structure         | 12.653           |
| 5                        | Cognitive Cues           | 12.121           |
| 6                        | Writing Style            | 11.901           |
| 7                        | Social Cues              | 11.268           |
| 8                        | Reviewer Experience      | 11.020           |
| 9                        | Psychological Distancing | 10.931           |
| 10                       | Perceptual Cues          | 8.221            |
| 11                       | Motion Words             | 8.105            |
| 12                       | Reviewer Reputation      | 7.211            |
| 13                       | Review Star Rating       | 2.074            |
| 14                       | Review Extremity         | 1.795            |
| 15                       | Number of Photos         | 0.402            |
| 16                       | Reviewer Location        | 0.154            |

classification models to improve their prediction performance by (1) splitting data into three partitions: training (model-building), validating, and testing datasets, (2) validating the performance of the trained (initially built) model, and (3) eventually suggesting the best-performing model with the testing dataset (Shah, 2017). This study applied seven different supervised ML-based classification techniques with three types of fake review characteristics (i.e. 16 predictors for fake reviews) from Yelp.com online restaurant reviews. Using our data based on their model performance measures, we found that the supervised ML-based classification technique 'random forest' is the best prediction model for fake review detection. This finding implies that random forest can be one of the best classifiers to predict fake reviews in the hospitality context. In addition, the relative importance test of fake review features reveals that each determinant may play a different role in detecting fake reviews, although prior fake review studies demonstrated that those features significantly influence fake reviews.

Thus, applying an ML-based classification model with various review features (i.e. review-, reviewer-, and linguistic characteristics) is methodologically and practically crucial for detecting fake reviews in the hospitality context. These findings can contribute to the field and can be brought into dialogue with work done by extant studies (e.g. Martinez-Torres and Toral, 2019). Theoretical contributions and practical implications are discussed in the following sections.

### *Theoretical implications*

This study has the following theoretical implications. First, to the best of our knowledge, the present study is one of the first attempts in the hospitality service context to integrate the possible factors used in detecting fake reviews suggested by various extant studies. The study empirically compares seven supervised ML-based classification models for their performance with large-sized data. Since Yoo and Gretzel (2009) mentioned the importance of comparing deceptive and truthful reviews, the field of hospitality service has lamented the fact that the majority of studies on fake review detection applying advanced classification techniques such as AI-based ML (with the fake review data from hospitality fields) has been published in non-hospitality service journals. Our literature review also found that most previous studies on detecting fake reviews applying AI-based ML algorithms have been published in non-hospitality or non-service journals (e.g. computer science, IS, and management journals). We found only a handful of studies on fake review detection or the factors influencing the occurrence of fake reviews that have been recently published in the hospitality service journals (e.g. Akhtar et al., 2019; Li et al., 2020; Martinez-Torres and Toral, 2019). In addition to the findings from these studies, the present study contributes to the hospitality literature by offering a complementary research method to analyze online reviews and develop a better prediction or classification model using advanced techniques. Therefore, the current study sheds light on the literature on the broad hospitality and service disciplines and specifically on fake online reviews and their detection methods in the hospitality field.

Second, the present study not only includes review-related attributes but also combines various attributes in terms of reviewer-related attributes and linguistic cues, while previous studies have focused only on review-related attributes (e.g. Martinez-Torres and Toral, 2019) in detecting fake reviews. Having a more comprehensive list of possible

determinants, this study can be considered more effective and robust. In particular, this study contributes to the literature by investigating both linguistic and non-linguistic features, as the inclusion of non-linguistic features could substantially enhance the prediction performance of fake online review detection methods (Zhang et al., 2016). Moreover, for the literature on the relative importance of various fake review determinants, there is still much to learn in the service industry as the extant literature has not provided many details about which review attribute (or feature) plays a more influencing role in fake review detection. In fact, prior studies on fake reviews have mainly focused on whether or not each of these features has a significant impact on the probability of being fake reviews, leaving room to investigate this issue further by applying an AI-based supervised ML approach. Thus, this study contributes to the literature on fake online review detection by expanding fake online review studies into more specific levels.

Third, this study contributes to the methodological approaches of the literature on online reviews in general. The study integrated business intelligence and machine learning procedures to develop and improve the fake review detection models through the following processes of (1) collecting data with web-scraping, (2) pre-processing data, (3) measuring linguistic features by applying a text mining tool, (4) training (i.e. initially building a detection model for performance validation) seven supervised classification models with different ML algorithms, (5) validating the trained models to compare their relative performance measures, and (6) examining the relative importance of 16 possible fake review predictors with the best performing model. Although we are not necessarily suggesting that the random forest algorithm should consistently perform the best across other ML-based classification techniques as its relative performance may vary based on datasets, our state-of-the-art methodological approach of handling online review data will contribute to the methodologies of future studies that conduct similar analyses with a large online review dataset in diverse contexts beyond the hospitality sector, such as electronic (online or mobile) commerce sites, online recommendation platforms, and sharing economy platforms.

### *Practical implications*

The present study contributes to practice as well. First, this study developed an enhanced fake review detection method by employing seven supervised ML algorithms for online review platforms. Findings of this study identified significant predictive features (i.e. linguistic, review-, and reviewer-related attributes) influencing fake review detection. The performance of seven classification (fake review detection) models was evaluated and compared to identify the most effective model. Our approach can help practitioners increase the credibility, transparency, and reliability of their online- or mobile-review systems for potential customers' purchase decision making, since customer reviews are considered more and more important not only in e-commerce settings but also in mobile- and omnichannel-commerce (Flavián et al., 2021) and online recommendation settings (Wu et al., 2020). In particular, such fake review detecting models with AI-based ML technologies could be beneficial for managers of online recommendation platforms wishing to improve their credibility. A recent industry report (Escobar, 2021) has suggested that the industry should flag suspect reviews and improve online review

verification or fake review detection systems for business owners as well as consumers, in the same line as our study's main research findings and contributions.

Second, this study contributes to increasing online market transparency and credibility. Our findings could help hospitality service providers, managers of online review platforms, and social media/communication marketers create and implement new fake review evaluation mechanisms. In the hospitality industry, fake reviews can critically reduce consumers' trust toward service providers, online review platforms, and market transparency, leading to customer attrition (Escobar, 2021; Salomon, 2019). Accordingly, practitioners can develop new management strategies based on such mechanisms. For example, according to our research findings, some attributes such as time distance (as a reviewer-related attribute), affective cues (as a linguistic attribute), and review depth (as a review-related attribute), among others, could be given more attention to reducing the probability of the occurrence of fake reviews, thus increasing the credibility and transparency of the review platform. More specifically, managers need to pay more attention to reviews (1) posted by reviewers who have only recently signed up to review, (2) are more emotional, and (3) that use fewer words (Li et al., 2020).

Lastly, the present study contributes to customers making better decisions by removing possible fake reviews and decreasing the possibility of information overload and online deception. That is, customers could recognize possible fake online reviews with relative ease by applying our ML models, and consumers' needs in finding authentic restaurants could be identified automatically via our predictive features. Consequently, consumers could make better decisions more efficiently and effectively using the ML-based classification (or detection) model online review platforms provide.

### *Limitations and future research*

This study has several limitations and can be used to provide suggestions for future research. First, the dataset of the present study was restricted to Yelp's 'Top 100 Places to Eat for 2017 in the U.S.' list, which could be problematic for generalizing our predictive model to other online review platforms targeted at other contexts and countries. Some U.S. restaurants may not be selected in the top 100 list, or a restaurant may not be listed in different years, and non-U.S. restaurants are excluded from the top 100 list. Second, although our determinants were adopted from extant studies, future research may extract other linguistic features by employing the natural language processing (NLP) technique to increase further predictive accuracies, such as making further investigations reviewing misspelling (Shan et al., 2021), linguistic dimensions (Plotkina et al., 2020), understandability (Banerjee et al., 2015), lexical validity (i.e. % of misspelling), and linguistic redundancy (Shan et al., 2021). Third, this study trained and validated the ML-algorithms with an assumption that the true-positive (fake) and true-negative (genuine) review screening processed by Yelp's own algorithm is 100% accurate, which cannot be verified unless Yelp checks with individual reviewers of the sampled data about whether or not the reviews screened as fake reviews are actually fake. We believe that this limitation cannot easily be addressed by future studies with the use of the Yelp.com dataset. This issue, however, should be addressed in future research by using advanced methods such as deep learning skills with new datasets. Fourth, linguistic cues were measured in this study using LIWC2015, which may exclude all the related

linguistic cues. Although LIWC2015 has been widely used in investigating fake reviews (e.g. Hancock et al., 2007; Li et al., 2020), future studies could conduct cross-validation of linguistic cues to increase the validity of the measurements. Lastly, while we believe that our suggested procedures and findings on fake review detection will contribute to the literature on online reviews, fake or manipulated reviews, which AI applications can also generate, are only one of the many aspects of unethical behaviors by consumers, businesses, and hackers. As more unethical behaviors can arise with the technological advancement of AI and as more decisions (e.g. detecting genuine vs. fake online reviews) are made without human interruption, future research on ML-algorithm should focus on addressing ethical issues regarding AI-based technologies (Belk, 2021), customer services issues with AI-based service interactions (Belanche et al., 2020b), and the issue of attributions of responsibilities for the decisions made primarily by AI rather than humans (Belanche et al., 2020a).

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